HW_Assignment_4_Fall2026

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ENEL 573 Computer Networks: HW Assignment 4 (Fall 2026)

(Due on Nov 6th, at 11:59pm Mountain Time)

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Problem 1 (10 points):

An organization with a class B address needs at least 950 subnets. Find the subnet mask and the configuration of each subnetwork.

Subnet mask = 255.255.255.192 /26

$2^n \geq 950$ Host ID = 16 - n

$n = 10$ bits = 16 - 10

= 6 bits

Problem 2 (10 points):

Find the subnetwork address and the host id for the following:

IP address: 120.14.122.16

Mask: 255.255.192.0

subnet address: 120.14.64.0/18

host ID: 0.0.58.16

n=2

IP	01111000 00001111 01111010 00010000
Mask	11111111 11111111 11000000 00000000
NOT Mask	00000000 00000000 00111111 11111111
IP AND Mask	01111000 00001110 01000000 00000000
IP AND NOT Mask	00000000 00000000 00111010 00010000

Problem 3 (10 points):

Consider sending a 1300-byte datagram into a link that has a MTU of 500 bytes. Which of the following is true? Please justify your answer.

1. Three fragments are created with offsets 0, 500/8 1000/8
2. Three fragments are created with offsets 0, 480/8 and 960/8.
3. Three fragments are created, with offsets 0, 460/8 and 920/8.
4. None of the above

1300 - 20 = 1280 bytes

500 - 20 = 480 bytes

$\frac{1280}{480} = 2.\overline{6}$ → 480, 480, 320

$0 + 480 = 480$

$480 + 480 = 960$

$\frac{480}{8} = 60$ $\frac{320}{8} = 40$

Problem 4 (10 points):

Suppose an IP datagram is fragmented into 10 fragments, each with an independent loss probability of 1%. What is the probability of receiving the datagram correctly at the destination if (a) the datagram (each fragment) is transmitted once; (b) the datagram (each fragment) is transmitted twice?

a) $P = (1-p)^{10} = (1-0.01)^{10} = 0.9044$

b) $P = (1-p^2)^{10} = (1-0.01^2)^{10} = 0.999$

Problem 5 (10 points):

Suppose nodes A, B, and C each attach to the same broadcast LAN (through their adapters). If A sends thousands of IP datagrams to B with each encapsulating frame addressed to the MAC address of B, will C's adapter process these frames? If so, will C's adapter pass the IP datagrams in these frames to the network layer C? How would your answers change if A sends frames with the MAC broadcast address (i.e., ARP request frames)? Please justify your answers.

Yes, C's adapter processes them because every frame reaches each adapter. But C will read that its for B's MAC so it won't pass the network layer.

Yes, C's adapter both processes and passes to the network layer because every adapter accepts a broadcast address